

Closure and the Foundations of Data Science

Abstract—Data science is typically grounded in two foundational principles: statistical validity and computational feasibility. These principles explain how to reason from data and how to realize that reasoning with finite resources, but they position data science as a collection of inherited methods and implicitly assume that analytic actions do not influence the systems that generate future data. Modern data-scientific practice routinely violates this assumption. We introduce closure as a third foundational axis that characterizes the coupling between analytic processes and their environments. Closure exposes structural limits on inference under feedback, distinguishes observational from participatory regimes, and reveals when the assumptions underlying validity and feasibility can be sustained. Together, validity, feasibility, and closure define a three-pillar foundation and a corresponding data science manifold that locates analytic systems by their evidential grounding, computational realizability, and degree of environmental participation. This framework provides an endogenous account of disciplinary purpose and reorients data science around the design and stewardship of analytic systems whose epistemic trajectories remain in regions where reliable knowledge can be sustained.

Index Terms—data science, statistics, computational theory

I. INTRODUCTION

Data science remains overly centered on tools. Drawing largely from statistics and computational theory, the field tends to treat the *problem*, *data*, and even the *purpose* of scientific inquiry as fixed and externally provided. In doing so, data science implicitly casts its aims as derivative of the domains it serves, and frames the discipline as fundamentally observational—a set of methods applied to questions articulated elsewhere—rather than participatory in shaping the environments and purposes it engages.

This disciplinary stance limits data science. It masks the fact that data science activities inevitably feed back into the environments generating the data they model: choosing what to measure, how to represent it, which models to train, and how to deploy them all change the system under study. A more complete foundation for the field must therefore account for this bidirectional coupling.

We propose that these limitations can be resolved by reframing data science as part of the very environments it observes. This requires introducing a third foundational pillar—*closure*—that complements the existing pillars offered by statistics and computational theory. As shown in Figure 1, closure provides a missing third axis that organizes disparate extensions of the statistical–computational plane into a single coherent structure. Where statistical theory provides the grounds for validity and computational theory for feasibility, closure addresses the interactive dynamics between data science processes and the systems in which they operate.

The term closure is chosen to name whether the relation between analysis and environment can be treated as effectively closed with respect to the assumptions that sustain inference. Closure marks the extent to which those inferential conditions can be preserved, rather than rewritten, by analytic participation itself.

By incorporating closure, data science expands from a discipline focused on inference and computation to one grounded in the dynamics of its participation in environments and in the expanded geometry revealed when all three axes are considered. This expanded three-pillar foundation offers:

- epistemic coherence, enabling data science to stand as an independent discipline rather than a derivative extension of parent disciplines or application domains;
- a geometric interpretation of data science practice and its developmental aims, in terms of statistical validity, computational feasibility, and closure; and
- an endogenous, structural framing of disciplinary purpose that complements the methodological orientations of statistics and computation.

Along the way, we introduce the concepts of *observational data science*, *participatory data science*, the *data science manifold*, and *effective data science* as the organizing vocabulary of this expanded perspective.

Closure highlights a major structural gap in the foundations of data science and articulates a conceptual space in which future theoretical and methodological development can proceed. Our aim is not to present operational algorithms for closure measurement or management, but to identify closure as the foundational axis that unifies these partial constructs and motivates their integration into a coherent research agenda for the discipline.

The pattern we identify parallels the historical development of computer science. Complexity theory provided a structuring axis that unified foundations in mathematics and electrical engineering greatly helping to establish computer science as a coherent discipline. Complexity is substrate-agnostic, method-agnostic, and domain-agnostic: it applies across algorithms, languages, machines, and representations.

Closure plays a structurally analogous role. It operates across statistical methods and computational resource settings and captures a domain-independent phenomenon at least as general as feedback. We emphasize this through examples, illustrating that:

- statistics and computational theory alone cannot address the coupling between data science activities and the systems within which they occur, and
- together, the three axes—statistical, computational, and closure—anticipate and unify many of the concerns

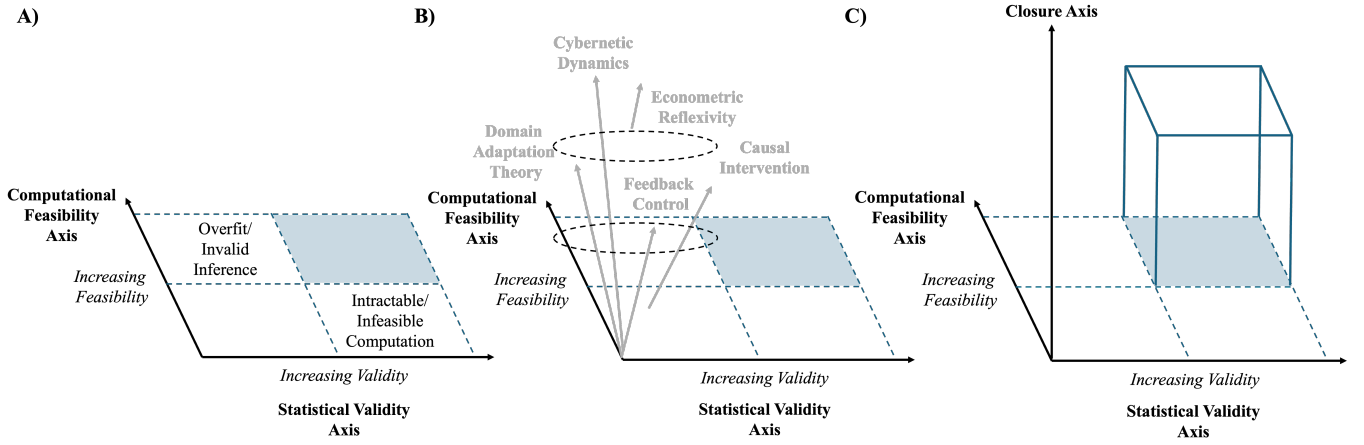


Fig. 1: From the statistical-computational plane to a unified three-pillar foundation. (A) shows the classical statistical-computational plane. (B) illustrates various partial or unaligned axes proposed to extend this plane. (C) shows closure as the structural third axis that integrates these concerns into a single geometric foundation.

that domain-specific research articulates, thereby refuting the common critique that data science is fundamentally derivative of domain practices.

The remainder of this paper proceeds as follows. We begin with historical context and a disciplinary analogy to computer science. We then review the two classical pillars of data science—statistical and computational theory—before defining closure and developing a geometric interpretation of data science grounded in the three-pillar foundation formed by statistical validity, computational feasibility, and closure. We discuss disciplinary and practical implications offered by this three-pillar foundation. We conclude with a discussion, synthesis, and remarks on future directions.

II. BACKGROUND AND MOTIVATION

A. Historical Emergence of Data Science

Although the use of data in science, administration, and engineering is centuries old, the idea of data science as a distinct intellectual and professional field is quite recent [1].

A common historical starting point is John Tukey’s 1962 essay “The Future of Data Analysis,” which anticipated a new, computation-rich practice of data analysis that would not fit neatly within classical mathematical statistics [2]. In the 1970s and 1980s, several authors began using the phrase “data science” in ways that foreshadow current usage. Peter Naur’s 1974 article *Concise Survey of Computer Methods* proposed “data science” as an alternative name for computer science, emphasizing data handling and modeling over hardware [3]. In parallel, C. F. Jeff Wu advocated in the mid-1980s and again in a 1997 lecture that statistics should effectively be renamed “data science,” to better reflect its central concern with data-driven inquiry and to shed narrower public stereotypes about statistics [4], [5].

Through the 1990s, workshops and conferences began to treat data science as a topic that combined statistics, data analysis, and computing; for example, the International Federation

of Classification Societies meeting in 1996 explicitly featured “data science” in this sense [6]. William Cleveland’s widely cited 2001 paper “Data Science: An Action Plan for Expanding the Technical Areas of the Field of Statistics” argued for a deliberately broadened field that integrated computational and data management concerns into statistics, effectively outlining an early “data science” curriculum [7].

The modern conception of data science as an independent discipline crystallized in the 2000s and 2010s, driven by the growth of large-scale digital data and the institutionalization of “data scientist” as a job title [8]. Davenport and Patil framed the data scientist as a new, high-impact role [9]. Professional bodies followed: the American Statistical Association renamed its “Statistical Learning and Data Mining” section to include “Data Science” in 2014 [10], and the European Association for Data Science (EuADS) was founded in 2015 after earlier European conferences on data analysis [11].

Recent scholarly and policy-oriented work explicitly treats data science as an emerging academic discipline. Some describe data science as emergent from an interdisciplinary integration of mathematics, statistics, computer science, and application domains [12]. Priestley analyzes data science as a “new mode of knowledge production” with institutional and epistemic consequences [13]. Lowrie focuses on the role of algorithms in shaping that knowledge production [14]. Fayyad proposes a “knowledge framework” for professional standards in data science, tracing the history of the term and arguing for a coherent body of knowledge that integrates statistics, computer science, and domain expertise [15]. Others consider the growing use of data in specific domains like sociology and medicine, and the consequences on the epistemologies of those domains [16]–[18]. Others yet still consider the intersection of data science and big data, and resulting consequences on empiricism and scientific theory development generally, and thereby epistemology itself (beyond data science) [19], [20]. These works, considered alongside the historical de-

velopment of data science, suggest that while data science has achieved institutional recognition, its disciplinary identity remains under-theorized: there is, as yet, no widely accepted account of what constitutes its own internal foundations.

Data science involves two distinct epistemic questions that must be carefully separated. First, there is the question of how particular analytical practices produce knowledge in specific contexts from data. Second, there is the question of what constitutes knowledge for data science as a discipline itself—what kinds of questions are legitimate, what counts as a contribution, and how the field justifies its own standards of inquiry.

This paper is primarily concerned with the second, discipline-level epistemic question. Unless otherwise stated, references to epistemic coherence, epistemic stability, or epistemological foundations should be understood as referring to the conditions under which data science functions as a coherent discipline, rather than to the correctness or performance of any particular analytical procedure in practice. As Desai et al. show in their survey [21], current attempts to address the second question, whether they be maximalist approaches [22] that “strive for sufficient conditions with detailed ontologies and methodological taxonomies” or minimalist approaches [23], [24] that “characterise early debates on the nature of data science”, are rooted in statistics, computational theory, or both. This paper represents a departure from this two-pillar foundation, arguing for a third pillar that provides the kind of structuring character that computational complexity theory provided computer science during its formative period as a discipline.

B. Computer Science as a Structuring Analogy

Computer science went through a phase where the practice of computing (building machines, writing code for specific domains) preceded a clear disciplinary theory. Denning’s classic overview “Computer Science: The Discipline” traces the field’s birth to the 1940s, when algorithm theory, mathematical logic, and the stored-program electronic computer converged [25]. Around the same time, theoretical work by Turing, Gödel, and others established the foundations of computability and formal languages, providing a mathematical account for what it means for a function to be computable by a machine [26]–[30]. Yet even with these foundational elements in place, the field lacked a theoretical axis that could organize its diverse mathematical and engineering components into a coherent disciplinary structure.

Computational complexity theory emerged during the 1960s–1980s as a systematic way to classify computational problems by the resources required to solve them—time, space, randomness—and to understand how these classes relate. [31]–[35]. Over time, the theory supplied computer science with its own internal questions and principles. Several influential authors emphasize that computational complexity theory helped define computer science as a discipline:

- Hopcroft’s 1987 article, “Computer Science: The Emergence of a Discipline”, explicitly credits the develop-

ment of computational complexity with sparking the imagination of computer scientists and helping establish theoretical computer science as an integral part of the field [36].

- Comer et al.’s 1989 article “Computing as a Discipline” identifies computational complexity theory as the component that “tells how to measure the time and space requirements” of algorithms and connects problem size to resource requirements, thereby organizing much of the discipline’s theoretical landscape [37].
- Surveys of theoretical computer science and its contributions (e.g., Savage [38]; Hartmanis [39]; Fortnow [40]) repeatedly present computational complexity theory as central to understanding the limits and capabilities of computation, which in turn distinguishes computer science from both mathematics and electrical engineering.

Taken together, these accounts suggest that computational complexity theory played a decisive role for computer science in resolving a condition similar to what data science faces today, where epistemic standards are inherited from parent disciplines, making the emerging discipline vulnerable to reduction into parent disciplines at precisely the historical moment when it is seeking to establish itself as an independent intellectual field.

Computational complexity theory did so not simply by improving methods, but by creating an internal epistemic structure that organized how knowledge in the field could be generated, compared, and delimited. In particular, computational complexity theory:

- introduced a domain-agnostic organizing variable—resource-boundedness—that allowed problems, methods, and results to be located within a shared conceptual space independent of hardware or application domain;
- generated internally owned questions through which the field could structure its own research agenda (e.g., P vs. NP, class separations, and reductions);
- stabilized internal standards for classifying problems, so that distinctions among forms of difficulty became epistemic properties of problem classes rather than contingent engineering judgments; and
- articulated principled limits on what kinds of computational questions could coherently be posed or resolved, thereby marking stable boundaries for legitimate inquiry.

Together, these contributions provided computer science with an internally coherent way to locate problems, evaluate results, and stabilize standards of knowledge across time. This kind of epistemic coherence is not static: it must be sustained. Epistemic stability, in this sense, is the condition that allows a field to preserve coherence as it grows, accumulates results, and trains new practitioners.

Drawing on this model, we adopt the following criteria for when a concept can function as a structuring axis for the epistemic stability of a discipline.

Definition 1 (Criteria for Epistemic Structuring). *A concept can function as a structuring axis for the epistemic stability*

of a discipline if it serves to:

- 1) *define a domain-agnostic organizing principle that allows problems, methods, and results to be located within a shared conceptual space for the field;*
- 2) *articulate intrinsic structural constraints that determine the scope of phenomena, relationships, and questions the field can coherently take as objects of inquiry;*
- 3) *provide stable, internally grounded standards by which the discipline evaluates whether particular contributions advance, refine, or reorganize its body of knowledge.*

If data science is understood as occupying a similar historical position to early computer science, then a central question becomes whether its inherited foundations from statistical theory and computational theory are sufficient to satisfy the conditions required for epistemic structuring. If they are, then the apparent epistemic gap in data science should, in principle, be resolvable through continued synthesis of these two traditions. However, even such sufficiency would not resolve the field's vulnerability to reduction into its parent disciplines or its dependence on external domains for purpose and evaluative meaning, leaving data science positioned primarily as a collection of methodologies rather than as a discipline with internally grounded epistemic commitments. If, on the other hand, these foundations do not meet the criteria for epistemic structuring, then a genuine gap in the internal epistemology of data science exists, even when treated as a collection of methodologies, and it becomes critical for the discipline to develop or identify additional foundational principles capable of stabilizing its identity as a coherent knowledge-producing field.

III. LIMITS OF THE TWO-PILLAR FOUNDATION

Data science inherits its core technical foundations from two mature theoretical traditions: statistical theory and computational theory. Together, these traditions supply discipline-level epistemic functions of statistical validity and computational feasibility. They specify what can be inferred from data and how such inference can be carried out within finite resource constraints. Despite their apparent functional role in data science, whether they can serve as satisficing epistemically structuring axes must be examined.

Statistical validity provides the analytical language of the field. It defines what constitutes valid inference, how uncertainty is represented, and how empirical claims are justified (Criterion 1). Concepts such as estimation, hypothesis testing, generalization, and identifiability establish principled boundaries between signal and noise (Criterion 2). It supplies internally coherent standards—such as consistency, unbiasedness, efficiency, and robustness—by which the field determines whether new methods meaningfully advance, refine, or reorganize its knowledge of valid inference (Criterion 3). At this high level, statistical validity appears to satisfy the epistemic structuring criteria.

Computational feasibility provides a complementary foundation. It specifies what types of inference are realizable

in practice, given limits on time, memory, and algorithmic structure (Criterion 1). Resource bounds, algorithmic complexity, optimization limits, and numerical stability constraints determine what can be feasibly computed (Criterion 2). It establishes stable evaluative criteria—such as tractability, scalability, optimality, and provable performance guarantees—through which contributions are judged to extend, sharpen, or restructure the discipline's understanding of feasible computation (Criterion 3). Again, at this level of abstraction, computational feasibility appears to satisfy the epistemic structuring criteria. But this apparent sufficiency depends critically on how the practice of data science itself is understood.

The canonical assumptions of statistical validity are that data are generated exogenously and that the inferential task can be specified without the act of analysis altering the data-generating process or the conditions under which inference is defined. The canonical assumptions of computational feasibility are that the problem formulation, objective function, and success criteria are externally supplied and remain stable as computation proceeds. Neither, in their canonical treatments, provide an *explicit account of the purpose, value, or consequences of analysis* in terms of its use, or of how *analytic actions influence the environments that generate the analyzed data*.

To be epistemically structuring according to the criteria, statistical validity and computational feasibility first require the meaning of analysis to be given, fixed, or otherwise external. And even then, they are sufficient only under the degenerate condition that data science is purely observational. In both cases, the characteristic of modern data science directly challenges these assumptions: in practice, data scientists commonly derive meaning for and through analysis to create value in the contexts where that analysis is used. Therefore, even at this coarse level, statistical validity and computational feasibility are only conditionally sufficient as epistemic structuring axes for the discipline of data science.

Proposition 1 (Conditional Sufficiency of the Two-Pillar Foundation). *The capacity of statistical validity and computational feasibility to function as epistemically structuring axes for data science depends on two limiting conditions:*

- 1) **Externalized Purpose Condition.** *The purpose, value, and consequences of analysis are treated as external to the discipline of data science and are not themselves objects of accountability within the field.*
- 2) **Non-Participatory Condition.** *The analytic process, and the actions it informs, are assumed not to alter the systems that generate the data on which analysis is performed.*

Thus, the two-pillar foundation does not satisfy the epistemic structuring criteria in systematic ways, and, to the extent it represents data science's current foundation, it leaves the field with systematic epistemic gaps.

- 1) Failure of a domain-agnostic organizing principle (Criterion 1). Neither uncertainty nor resource constraints can coherently organize the field once analytic actions alter

data-generating dynamics, leaving the field without an axis to locate its own central problems under feedback and reflexivity.

- 2) Failure of principled internal scope (Criterion 2). In the absence of a theory of analyst–environment coupling, data science cannot determine which kinds of relationships, interventions, or feedback structures are coherent objects of inquiry—beyond the purely observational case of “no coupling”.
- 3) Failure of stable standard of contribution (Criterion 3). Improvements in statistical validity or computational feasibility alone cannot function as stable markers of disciplinary progress without a concept of the analyst–environment couplings where such improvements hold.

To restore epistemic stability under participatory conditions, data science requires a third structuring axis—one that explicitly organizes analyst–environment coupling. The next section proposes such an axis, develops its minimal theoretical structure, and demonstrates in principle how it resolves the systematic gaps left by statistical validity and computational feasibility. We then examine how this axis, taken together with validity and feasibility, forms a three-pillar foundation that can be represented geometrically, clarifying how different data-scientific settings occupy distinct regions of this space. Subsequent sections use this foundation to analyze implications for the discipline and for practice, showing how the resulting framework provides coherent standards for inquiry, contribution, and epistemic stability.

IV. CLOSURE

For data science, a source of epistemic instability arises from the fact that analytic processes operate within environments that may themselves change in response to analytic actions. To make this interaction analytically explicit, data science requires a structural axis that we term *closure* to characterize how strongly analytic actions shape the future data-generating environment. At a high level:

- *Low closure* describes settings where analytic actions have negligible effects on the environment’s evolution and thus on future data.
- *High closure* describes settings where analytic actions materially reshape the environment, altering the conditions under which subsequent data are produced.

Closure is not a statement about the correctness or feasibility of inference. It is a property of the *coupling* between an analytic process and its environment—specifically, how the consequences of analytic actions propagate into the future data-generating process.

Definition 2 (Closure). *Closure characterizes how strongly analytic actions reshape the conditions under which future data are generated. Low closure indicates that analytic actions have negligible effects on the environment; high closure indicates that analytic actions materially alter the future data-generating process.*

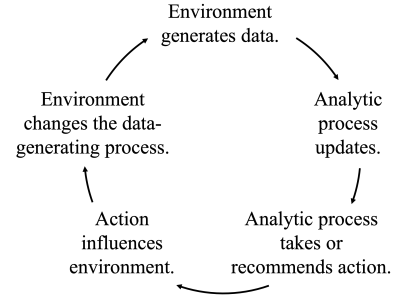


Fig. 2: The analytic process characteristic of modern data science that engenders non-zero closure.

This section proceeds as follows. We begin by developing a process-level account of closure, formalizing the interaction loop between an analytic process and its environment to make explicit the point at which analytic actions shape future data conditions. We then introduce axioms that any closure measure must satisfy, establishing the structural properties that distinguish closure from statistical validity and computational feasibility. Building on this foundation, we identify qualitative limits that arise in high-closure regimes, showing how strong analyst–environment coupling can undermine the assumptions necessary for reliable inference. We next situate closure within adjacent theoretical traditions—such as control, cybernetics, game theory, and performative prediction—to demonstrate how these areas implicitly engage the same axis and why their mechanisms cohere under the concept of closure. Finally, we show how the resulting framework directly addresses the epistemic gaps identified in Section III, thereby positioning closure as a foundational axis capable of restoring epistemic stability to data science.

A. Process-Level View

To make closure more precise, we describe a minimal interaction loop between an analytic process and its environment, illustrated in Figure 2. This loop is intentionally agnostic to modeling paradigm, implementation details, or the presence of human analysts; the analytic process may be a statistical model, a hybrid human-machine workflow, or any agent that performs inference and produces actions.

At each time step t :

- 1) **Environment.** The environment occupies some state E_t . This state determines how data are generated and how the environment will react to actions.
- 2) **Data.** The environment produces observations D_t , sampled from a distribution determined by the current environment state:

$$D_t \sim P(\cdot | E_t).$$

- 3) **Analytic Process.** The analytic process occupies an internal state M_t , which encodes whatever information or structure it uses for inference (e.g., beliefs, model parameters, heuristics, or human-supplied judgments). Using M_t and the observed data D_t , it produces an action.

- 4) **Action.** The analytic process outputs an action A_t . Actions include predictions, decisions, recommendations, control signals, or other interventions.
- 5) **Environment Evolution.** The environment transitions to its next state according to a transition function

$$E_{t+1} = f(E_t, A_t),$$

where the dependence on A_t captures how analytic actions shape future conditions.

- 6) **Analytic Update.** The analytic process may update its internal state from M_t to M_{t+1} using the new observations:

$$M_{t+1} = g(M_t, D_{t+1}).$$

Closure concerns the strength of the dependence E_{t+1} on A_t . Equivalently, it measures how much analytic actions structure the conditions that will generate future data.

B. Closure Measures and Axioms

We do not commit to a single numerical measure of closure. Any closure measure C is acceptable provided it obeys certain structural axioms. These axioms capture the essential features of how analytic actions shape future data-generating conditions.

Axiom 1 (Zero-closure). *Closure is zero if and only if analytic actions have no effect on the environment’s evolution that is relevant to future data. Formally,*

$$f(E_t, A_t) = f(E_t) \text{ for all } t \iff C = 0.$$

When $C = 0$, the environment’s data-generating process evolves independently of the analytic process on the timescale of interest. This is the purely observational setting: the analytic process may depend on the environment, but the environment does not depend on the actions produced by analysis.

Axiom 2 (Monotonicity in coupling strength). *If the coupling between analytic actions and the environment increases—while holding fixed the representation of states and actions—then closure should not decrease. More precisely, if a modification to the analytic process increases the dependence of*

$$E_{t+1} = f(E_t, A_t)$$

on the action A_t , or increases the portion of the environment whose evolution depends on those actions, then the closure measure C should be weakly larger. This axiom formalizes the idea that stronger action-dependent environment evolution corresponds to a higher degree of closure.

Axiom 3 (Separation from validity and feasibility). *Closure is a structural property of how analytic actions shape future environmental states. Therefore:*

- *Improving or degrading the statistical validity of the analytic process—without changing how actions affect environment evolution—must not change closure.* (Better or worse inference with the same action-dependent environment dynamics leaves closure unchanged.)

- *Improving or degrading the computational feasibility of the analytic process—without changing the action-environment coupling—must not change closure.* (Faster or slower inference with the same pattern of actions and environmental response leaves closure unchanged.)

This separation does not imply that closure, validity, and feasibility are unrelated in practice. Rather, it ensures that closure, as a foundational axis, remains well-defined and does not conflate environmental coupling with analytic performance. Their interaction emerges in the geometry defined by these three axes, not by embedding one axis inside another.

C. Limits of Inference Under High Closure

Even without specifying a particular formula for closure, we can identify qualitative limits that arise when closure becomes sufficiently high. These limits do not concern analytic correctness, computational feasibility, or modeling choices. Instead, they reflect structural boundary conditions: settings in which the environment’s dependence on analytic actions becomes incompatible with maintaining the assumptions required for reliable inference.

We provide two such limits, presented as informal theorems. Each shows how, at high closure, the very conditions that make analysis possible can erode faster than an analytic process—human or algorithmic—can track.

Theorem 1 (Instability of Assumptions Under Rapid Environment Change). *When analytic actions reshape the environment on a timescale comparable to or faster than data can be collected and processed, there may be no nontrivial window in which assumptions calibrated on past data remain valid for future inference. In such regimes, stable relationships between observations and underlying phenomena may be structurally unavailable.*

Example 1 (Adversarial Drift Driven by Action Response). In large-scale fraud detection, models are continuously updated from recent transaction streams. At moderate closure, fraud patterns drift slowly enough for recent data to support short-term predictive assumptions. But when attackers automate their adaptation to model behavior—probing and shifting tactics within minutes—analytic actions (blocking patterns, flagging flows) reshape the threat landscape faster than new data can be aggregated and learned from. The environment’s dynamics outpace inference, leaving no stable distribution to model.

Example 2 (Interventions That Reshape Dynamics Faster Than Observation). In closed-loop ecological management, interventions (e.g., targeted species removal and nutrient adjustments) are taken based on model predictions. If interventions significantly alter population dynamics or biogeochemical cycles faster than updated measurements can be acquired, the system’s response invalidates the assumptions used to design the intervention itself. The environment becomes a moving target whose governing dynamics shift faster than observational updates permit.

Theorem 2 (Incompatibility of Strong Action Effects with Long-Term Invariance). *In some regimes, maintaining strong, persistent action effects is fundamentally at odds with preserving a stable data-generating environment. When analytic actions continually reshape the environment, the notion of an invariant or representative distribution for long-term analysis ceases to be well-defined.*

Example 3 (Continuous Optimization That Eliminates Baseline Conditions). In city-wide traffic routing systems, real-time navigation guidance intentionally induces large behavioral changes: rerouting substantial fractions of drivers to optimize flow. Effective interventions eliminate the very congestion patterns being modeled. Over extended periods, the environment settles into traffic dynamics created by the routing logic itself, preventing any long-run invariant distribution that reflects pre-intervention behavior.

Example 4 (Closed-Loop Exploration that Rewrites the Search Landscape). In robotic materials discovery, closed-loop algorithms iteratively choose experimental conditions to maximize promising outcomes. When these choices systematically reshape the underlying search landscape—for example, by altering chemical equilibria or reaction pathways—there may be no stable background distribution of outcomes. The optimization policy continually redefines the environment, precluding any long-term invariant model of system behavior.

These limits illustrate that high closure describes more than increased complexity: it marks settings in which essential conditions for data science—stability, representativeness, invariance—can fail categorically. At sufficiently high closure, no refinement of models, inference techniques, or computational resources can restore these conditions. Closure thus delineates the boundary between settings where reliable data-scientific practice is attainable in principle and settings where the environment’s dependence on analytic actions renders such reliability structurally out of reach.

D. Relation to Existing Theoretical Traditions

Closure is introduced here as a third epistemic axis, but the phenomena it formalizes are not new. As with statistical validity and computational feasibility—each of which gathers diverse theoretical traditions under a common conceptual role—closure brings coherence to a set of ideas that already appear across many fields. What is new is not the component theories themselves, but their synthesis into a foundational axis defined with the epistemic needs of data science at its center.

A concept contributes to the closure if it satisfies all three of the following conditions:

- 1) *Analyst participation is linked to future data conditions.* The construct must formalize a mechanism by which analytic actions, decisions, or outputs change the subsequent data-generating process. This includes, e.g., feedback loops, adaptive sampling, reflexive behavior, intervention-induced structural changes, and policy-dependent drift. This excludes purely observational notions of nonstationarity.

TABLE I: Conceptual traditions whose constructs satisfy the criteria for contributing to the closure axis.

Field / Tradition	How the Construct Contributes
Control Theory	Provides quantitative mechanisms showing how actions propagate through system dynamics and identifies stability boundaries (closure thresholds) [41], [42].
Cybernetics and Adaptive Systems	Models recursive action–environment adjustment and clarifies when such adjustments preserve or erode the conditions enabling reliable inference [43], [44].
Dynamical Systems	Characterizes regime shifts caused by analytic perturbations and delineates closed (stable) vs. open (unstable or bifurcating) regimes [45], [46].
Game Theory and Mechanism Design	Formalizes behavioral adaptation in response to analytic interventions, showing when reflexive responses preserve or invalidate model semantics [47], [48].
Causal Intervention Theory	Represents how deliberate analytic actions modify generative structure and identifies when such modifications preserve identifiable causal semantics [49], [50].
Reinforcement Learning and Online Learning	Models settings where data become endogenous to analytic policies; contributes criteria such as regret or convergence bounds that mark closure-preserving regimes [51], [52].
Performative Prediction	Models prediction-driven distribution shift and quantifies when predictive systems converge to or diverge from closure-preserving equilibria [53].

- 2) *The construct under-specifies how those changes affect the assumptions that underwrite inference.* It must characterize the way environment response modify (or preserve), e.g., identifiability conditions, model validity assumptions, independence or stability assumptions, predictive semantics, and performance guarantees (statistical or computational). It is not enough that the environment changes; the construct must connect those changes to the epistemic constraints relevant for inference.
- 3) *The construct provides a concept for the boundary between closure and non-closure.* The construct must make contributions with reference to either a quantitative measure that tracks approach to a closure threshold (e.g., drift rate, stability margin) or a structural distinction that demarcates lower or upper limits on closure (e.g., endogenous vs. exogenous, stable vs. unstable). This boundary-locating function makes the construct epistemically integrable into the axis.

In short, a concept contributes to the closure axis when it formalizes how analytic actions reshape future data conditions, how those reshaped conditions affect the assumptions required for inference, and where the boundary lies between regimes that preserve or destabilize those assumptions. Table I situates several representative examples to clarify how closure draws from, rather than replaces, these existing fields and traditions.

In this respect, closure plays a structurally analogous role to computational complexity. Before computational complexity theory, multiple notions of computational difficulty existed—running time, memory usage, tape length, circuit depth—but

no unifying axis organized them. Computational complexity did not replace those constructs; it revealed the underlying structural dimension along which they aligned. Similarly, closure brings data-scientific alignment to notions of feedback, drift, reflexivity, and intervention: it identifies the common axis implicit to existing theories and formalizes it with the epistemic aims of data science as its foundation.

This aligns closure with the other two foundational pillars. Validity draws on probability, statistical decision theory, asymptotic arguments, and causal logic; feasibility draws on algorithms, complexity theory, numerical analysis, and systems design. These heterogeneous traditions cohere in service of data science because the discipline interprets them as characterizing whether inference *should be trusted* (validity) and whether it *can be carried out* (feasibility). Closure gathers equally heterogeneous constructs from machine learning, control, cybernetics, and adaptive decision-making, but does so in service of the third foundational question: whether inference *remains informative* as it interacts with the system that generates future data.

E. Closure and the Sufficiency Criteria for Epistemic Stability

The closure axis was introduced to address the systematic epistemic gaps left by the two-pillar foundation of validity and feasibility identified in Section III. We now highlight how closure satisfies these gaps.

- 1) Domain-agnostic organizing principle (Criterion 1). Validity and feasibility cannot organize the field once analytic actions alter the dynamics of data generation. Closure satisfies this gap by providing a domain-agnostic axis along which all data-scientific settings can be located: from purely observational settings with zero closure to tightly coupled, closed-loop systems with high closure.
- 2) Principled internal scope (Criterion 2). Without a theory of analyst–environment coupling, data science lacks intrinsic criteria for determining which phenomena and relationships are coherent objects of inquiry. Closure satisfies this limitation by specifying the structural conditions under which stable inference is possible. As demonstrated by the limits of inference under high closure, certain problem types are epistemically unavailable when analytic participation reshapes the environment too rapidly or too extensively.
- 3) Stable standard of contribution (Criterion 3). Improvements in validity or feasibility alone do not reliably indicate disciplinary progress without a concept of the closure regimes to which they apply. Closure satisfies this third limitation by grounding evaluative standards in structurally explicit conditions: advances in validity or feasibility are meaningful only when the assumptions they refine or extend are stable under the relevant degree of analytic participation.

These considerations address the non-participatory condition limiting the sufficiency of the two-pillar foundation. The externalized purpose condition—concerning how data science

positions itself relative to the purposes and contexts within which it operates—is taken up later in the discussion in Section VIII. For now, it is sufficient to note that closure places analytic processes within the environments they act upon, enabling a later account of purpose and context that builds on, rather than precedes or parallels, the technical foundation established here.

V. FROM CLOSURE TO THE THREE-PILLAR FOUNDATION

With closure addressing the epistemic gaps left by the two-pillar foundation, we obtain a three-pillar foundation for data science. *Validity* characterizes the evidential grounding of inference; *feasibility* characterizes the computational resources required to realize it; and *closure* characterizes how analytic actions shape the environment that generates future data. Together, these three axes supply a structural space in which analytic tasks, methods, and contributions can be coherently located.

A. The Data Science Manifold

These three axes define a conceptual manifold: a space in which every analytic setting occupies a position determined by its validity, feasibility, and closure properties. The manifold is not a metric geometry but a coordinating framework—analogue to the role played by computational complexity theory in computer science—within which heterogeneous analytical situations can be organized without appeal to external disciplinary epistemologies.

Closure provides a language to distinguish two modes of practice:

- **Observational data science**, where analytic actions do not meaningfully alter the data-generating environment and the classical statistical–computational plane suffices; and
- **Participatory data science**, where analytic outputs influence future data conditions and must therefore be incorporated into the epistemic structure of the task.

Most modern analytic systems operate in the participatory regime. The manifold provides a unified representation for both.

Definition 3 (Data Science Manifold). *The data science manifold is the three-dimensional conceptual space spanned by statistical validity, computational feasibility, and closure. Each point represents data-scientific activity by its evidential grounding, its computational complexity, and the degree of coupling between analysis and environment.*

Within this manifold, illustrated in Figure 3, different regions describe the conditions under which inference can or cannot be sustained:

- 1) *The statistical–computational plane* (*closure* = 0). Purely observational settings where environment dynamics can be treated as exogenous. Classical statistical and algorithmic theories reside here, but this region is a degenerate slice from the perspective of the full manifold.

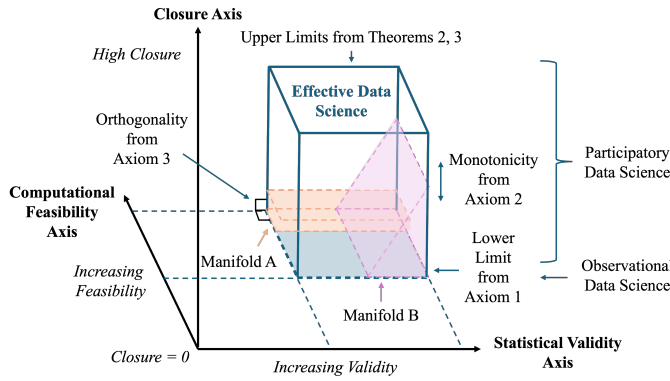


Fig. 3: Geometric view of effective data science. The larger rectangular prism shows the region of effective data science, with observational and participatory subregions. Example manifolds are shown for two analytic paradigms: a robust-statistics workflow with limited computation (rectangular prism with low closure) and an online learning system with continual retraining (triangular prism extending further along the closure axis). These illustrate how different workflows occupy distinct feasible regions defined by validity, feasibility, and closure.

- 2) *Intermediate closure.* Settings where analytic actions influence the environment, but not so rapidly that inference becomes unstable. Here, all three pillars jointly constrain what analyses are possible: evidence remains informative, computation remains tractable, and analyst-environment coupling remains manageable.
- 3) *High closure.* Beyond a structural threshold—identified in Section IV—environmental responses to analytic actions evolve faster than inference and computation can track. Validity, feasibility, and closure jointly mark this boundary: when any one of the pillars collapses, data-scientific inquiry cannot be sustained.

These regions are not normative categories; they reflect structural limits on inference.

B. Effective Data Science

A central implication of the manifold is the identification of a bounded region in which data-scientific reasoning is possible at all. We call this region *effective data science*.

Definition 4. *Effective data science denotes the region of the data science manifold in which inference, computation, and analyst-environment coupling are jointly compatible: evidence remains informative, computation remains feasible, and the degree of closure does not preclude stable assumptions for analysis.*

This region is the counterpart, for data science, to the tractable classes of computational complexity theory: it specifies *where the discipline can operate coherently*, and therefore where epistemic standards can be meaningfully applied.

Figure 3 depicts effective data science as a geometric object with subregions defining observational and participatory

data science. Inside the region of effective data science, two example manifolds are illustrated corresponding to different analytic paradigms. Manifold A represents a robust-statistics workflow with limited computational resources and no capacity for continual retraining. Geometrically, it appears as a rectangular prism that rises about the statistical-computational plane only to a low level of closure: validity spans a large region, feasibility is high, but coupling to the environment remains negligible. Manifold B represents an online learning system that continually retrains lightweight models with access to dynamically scalable compute resources. It has a larger computational feasibility region but spans a narrower band of statistical validity, reflected reduced robustness. Its geometry is a triangular prism that extends further along the closure axis: its maximal closure occurs at the highest statistical validity, and then slopes downward as validity and closure decline together. These example illustrate how different analytic paradigms carve out distinct feasible manifolds in the effective data science region and how their epistemic constraints are expressed geometrically.

C. The Role of the Three Pillars in Epistemic Stability

The three-pillar foundation satisfies the epistemic-structure criteria introduced in Section II:

- *Domain-agnostic organizing principle.* The manifold provides a shared conceptual space locating all analytic settings, from classical statistics to adaptive machine learning systems to high-interaction socio-technical loops.
- *Principled internal scope.* The structural interplay between validity, feasibility, and closure delineates which analytic relationships and interventions are coherent objects of inquiry. In particular, the limits of high closure mark intrinsic epistemic boundaries.
- *Stable standards of contribution.* Improvements in statistical validity or computational feasibility are meaningful only relative to the closure regimes in which they hold; closure is meaningful only relative to the validity and feasibility constraints it must respect. The epistemic stability of the field arises not from any pillar alone, but from their *interaction*.

Thus, it is the *three-pillar foundation as a whole*, not closure in isolation, that supplies data science with a coherent internal epistemology. Closure repairs the missing dimension; validity and feasibility anchor the resulting space; together they form the geometric and conceptual basis for a discipline capable of standing on its own epistemic grounds.

VI. IMPLICATIONS FOR THE DISCIPLINE

By supplying a domain-agnostic organizing principle and internal scope conditions, the three-pillar foundation reframes longstanding problems as structural rather than domain-specific and reveals new areas in which theory is required. It gives the discipline a unified way to understand when analytic systems remain within the region of effective data science—regardless of the scale or setting in which they operate.

A. Testability as a Structural Property

In conventional statistical learning, *testability* is treated as a dataset-centric notion: models have assurance because they have been tested on held-out samples assumed to represent future conditions [54]. This framing presumes a clean separation between model development and model deployment, and a stable target distribution against which generalization can be certified. But closure breaks this assumption [55].

Consider condition-based monitoring of machine health. At first, it appears to be a standard statistical task: infer latent health state from sensor data. But once those inferences trigger maintenance, inspection, recalibration, or sensor relocation, the intervention alters the subsequent data-generating process itself [56]. The resulting degradation of analytics is not primarily a behavioral matter of noise, drift, or inadequate data coverage; it arises because deployment reconstitutes the very system under analysis—a structural constraint affecting the data generating process. Under closure, a model may be validly tested in one regime and yet be untestable in the classical sense once its use helps produce another.

Closure shows that classical testability is a low-closure case. In condition-based monitoring, a model may be validated on historical sensor data, but once its outputs trigger maintenance or system changes, deployment alters the very process being evaluated. The issue is not ordinary drift but action-dependent change in the data-generating environment. High-closure systems are thus not untestable, but testability becomes local, closure-aware, and tied to whether evaluation remains informative over the relevant timescale.

B. Personalization, Validity, and the Feasibility of Governance

Recommender systems and personalized agents appear, at first glance, to be classical prediction problems: estimate user preferences and deliver individualized content. Yet these systems instantiate some of the strongest forms of closure in contemporary data science [57], [58]. Every recommendation alters user behavior; user behavior alters future data; and future data alter subsequent recommendations.

The three-pillar foundation suggests a structural trilemma for data-scientific governance: a system with strong personalization cannot, in the limit, simultaneously maintain:

- 1) *High personalization (high closure)*—agent-specific adaptation to user behavior;
- 2) *Statistically valid population-level evaluation*—classical testing, randomized experiments, or representative sampling;
- 3) *Computationally feasible governance*—monitoring, auditing, and constraint enforcement under realistic budgets.

Only two of these can coexist at a time. Strong personalization destroys population exchangeability; valid population testing requires suppressing personalization; feasible governance cannot compensate for the combinatorial explosion that arises when every user occupies a distinct analytic-environment coupling. This is a discipline-level analogue to complexity

classes: a sketch of a structural impossibility proof, not a contingent engineering difficulty.

C. AI Ethics as Implicit Closure Reasoning

In socio-technical settings—employment, credit, education, healthcare—data science operates within systems where human and institutional behavior adapt to analytic outputs. Predictions reshape opportunities; opportunities reshape data; data reshape future predictions. These settings exhibit high closure, though this has rarely been formalized.

AI ethics has long grappled with harms emerging from such feedback loops: disparate impact, representational bias, structural exclusion, runaway optimization dynamics [59]–[62]. These critiques often rely on extended causal narratives describing how models reshape the environment in which they operate. What appears as a moral or sociological critique is, at its core, implicit closure reasoning. Rather than connecting the epistemologies of ethics and AI directly to address these harms [63], AI can be treated as statistical-computational analytical objects with closure-like couplings to society, using closure as the rigorous, common conceptual ground for framing such discourse in a manner that supports the development of the disciplinary ethics of data science.

The manifold reveals why technical fairness metrics, defined solely on distributions and error rates, fail to capture these dynamics: they live on the statistical-computational plane and assume a closed-world model of inference. Ethical failures frequently arise not from error but from unacknowledged participation—from analytic actions that alter future data conditions in ways the two-pillar foundation cannot represent. That is, many ethical failures are not just misapplications of existing validity criteria, but failures to account for high-closure regimes.

VII. IMPLICATIONS FOR PRACTICE

A. Epistemic Trajectories

The data science manifold provides a coordinate system for describing the state of an analytic process at a given time: its evidential grounding (validity), its computational realizability (feasibility), and its degree of analyst-environment coupling (closure). Real systems do not occupy a single point in this space; they move.

Definition 5 (Epistemic Trajectory). *An epistemic trajectory is the time-indexed path traced by an analytic process through the data science manifold as it is deployed, updated, and allowed to interact with its environment. Each point on the trajectory represents the joint state of validity, feasibility, and closure at a given time.*

Once epistemic trajectories are taken as primary, core practices in data science become questions of trajectory management. Because these trajectories are shaped by modeling choices, update policies, and deployment contexts, practitioners must design not only models, but the motion those models will induce through the manifold. The manifold suggests

several concepts along which practitioners can reason and design, including:

- *Closure continuity.* The property that a system’s epistemic trajectory maintains a consistent relationship between its outputs and the user-generated evidence that informs future updates. In recommender systems, loss of closure continuity can occur when a model update forces the trajectory to re-establish personalization from a low-closure state.
- *Epistemic velocity.* The rate at which the system’s position in the manifold changes over time. High velocity indicates rapidly changing assumptions, models, or environments. In recommender systems, aggressive retraining and fast user adaptation increase velocity, demanding more intensive monitoring and governance.
- *Trajectory curvature.* How quickly the direction of motion changes. High curvature reflects regimes where small perturbations in policy or environment produce large qualitative changes in behavior (e.g., tipping points in engagement dynamics). Curvature is a proxy for sensitivity to design decisions. In recommender systems, trajectory curvature becomes visible when a small change in ranking policy, feedback weighting, or interface design sharply redirects user behavior, for example by pushing the system from diverse exploration into a narrow engagement loop.
- *Boundary proximity.* The distance from the current state to regions where one of the pillars fails: loss of validity (assumptions no longer hold), infeasibility (resource constraints are violated), or excessive closure (assumptions cannot be maintained at any reasonable update cadence). In recommender systems, boundary proximity is evident when the system approaches regimes where personalization degrades because user feedback becomes too sparse, retraining costs exceed operational limits, or interaction patterns shift so quickly that prior assumptions no longer support reliable updates.

Instrumenting systems to track these concepts is a practical implication of the three-pillar foundation. For recommender systems, this might include monitoring whether personalization behavior learned before a model update remains coherent afterward (closure continuity), how often model updates invalidate previous evaluation results (epistemic velocity), how sensitive engagement metrics are to small changes in recommendation policies (curvature), and maintaining explicit thresholds beyond which personalization is throttled or decoupled (boundary proximity). Having seen how trajectory characteristics shape system behavior in a concrete setting, we now turn to how these same principles reframe core data-scientific practices more broadly.

B. Reframing Core Practices

Epistemic trajectories provide a new object of practice. Data scientists are not custodians of static models evaluated against fixed distributions; they are designers and stewards of

epistemic trajectories—dynamic paths that depend on the joint behavior of validity, feasibility, and closure.

- *Model development and selection.* Under the two-pillar view, model selection seeks to maximize validity subject to feasibility on a fixed task. Under the three-pillar view, development determines an initial position and direction in the manifold. Decisions such as whether to adopt a static model, a periodically retrained model, or a fully online learner now determine the expected epistemic velocity and closure continuity of the resulting system. As the recommender case demonstrated, these design choices decide whether trajectories remain near the statistical-computational plane or immediately enter high-closure regimes.
- *Evaluation and experimentation.* Offline evaluation presumes that test data approximate future conditions. In participatory systems, this assumption is fragile: analytic actions reshape the data-generating environment. Evaluation therefore becomes a repeated sampling of the trajectory, not a one-time assessment of a static model. Experiments estimate local properties such as velocity, curvature, and boundary proximity, enabling practitioners to identify when assumptions are degrading and when a system is trending toward unstable closure regimes.
- *Monitoring and governance.* Operational monitoring traditionally focuses on performance and service metrics. The trajectory perspective augments this with structural indicators that track how quickly a system is moving through the manifold, how strongly it is coupled to user behavior, and how close it is to validity- or feasibility-breaking regions. Governance mechanisms—throttling personalization strength, enforcing exploration, inserting human review—can be understood as deliberate attempts to bend, slow, or redirect epistemic trajectories to keep them within the region of effective data science.
- *Documentation and professional judgment.* Documentation also shifts. It is no longer solely a record of model architecture and dataset properties, but of the intended trajectory: the assumptions under which the system should operate, the regions of the manifold it is designed to inhabit, and the safeguards meant to keep it there. Professional judgment becomes a matter of articulating and defending not only point-wise design choices, but the motion those choices will induce over time.

VIII. DISCUSSION

A. Closure, Value, and the Externalized Purpose Condition

Under the two-pillar foundation, data science is only epistemically stable if it admits what Section III called the Externalized Purpose Condition: domains supply the aims, costs, and utilities of analysis—*value*, put broadly—while validity and feasibility govern inference and computation. This arrangement works in purely observational settings, where analytic actions do not appreciably alter the systems that generate future data.

Non-zero closure disrupts this separation. In participatory settings, the values used in analysis shape actions taken, and those actions reshape the conditions under which future data are produced. Values are no longer external to inference; they help determine the closure regime and thus whether the assumptions required for validity and feasibility can be sustained over time.

Closure does not prescribe specific values—costs in fraud detection, engagement targets in recommenders, clinical risks—but it does clarify their structural roles. It at least distinguishes:

- **observational value**, where purpose guides evaluation but does not affect future data (zero closure), and
- **participatory value**, where purpose alters environment dynamics and hence the stability of inference itself (non-zero closure).

This distinction provides a path for the three-pillar foundation to resolve the Externalized Purpose Condition: purpose is in part structurally intrinsic to the joint behavior of validity, feasibility, and closure, rather than solely derivative of use.

B. Against Premature Formalization of Closure

Closure is introduced here at the same foundational level as statistical validity and computational feasibility. Neither of those pillars is defined by a single quantitative formalism; each is supported by multiple theoretical traditions—statistical inference, generalization theory, optimization, complexity, algorithmic design—without any expectation that they converge to a unified metric. Their role in the foundation is structural, not monolithic.

Closure should be treated in the same spirit. Formalizing it too early, or forcing it into a single measure, would collapse it into the specific fields from which it draws—such as performative prediction, control, reinforcement learning, and causal modeling—rather than allowing it to play the integrative role needed for data science. Closure is meant to organize these traditions, not to be reduced to any one of them.

A mature theory of closure will likely resemble the other pillars: a family of compatible frameworks connected by shared structural commitments. The axioms introduced here provide that structure; the diversity of formal expressions will emerge naturally as the field develops.

IX. CONCLUSION

This paper has argued that the inherited foundations of data science—statistical validity and computational feasibility—are necessary but not sufficient to secure the epistemic stability of the discipline. They provide rigorous accounts of how to infer from data and how to realize that inference within finite resources, but they do so only under the limiting assumptions that analysis does not influence the systems that generate future data and that the purpose of analysis can remain external to the discipline itself.

Closure introduces a third epistemic axis that captures the strength and consequences of analyst–environment coupling. It also provides a structural concept for data-scientific theories

of value and thereby purpose. Incorporating closure alongside statistical and computational foundations yields a three-pillar foundation and a geometry that locates analytic systems by their evidential grounding, computational realizability, and degree of participation in the environments they study.

Much remains to be developed. A mature theory of the three-pillar foundation would articulate principles to more rigorously offer the discipline what complexity theory offered computer science: a structural vocabulary for distinguishing what is possible, what fails for principled reasons, and what counts as genuine epistemic progress.

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